

SMART PROCTOR AI

Ensuring Academic Integrity through Real-Time Concentration Tracking



Presence Detection

Instantly detects if the candidate leaves the frame or if multiple faces appear during the exam.



Gaze Tracking

Monitors eye movement and iris landmarks to ensure visual focus remains directly on the screen.



Instant Alerts

Automatically dispatches SMTP email alerts to proctors with timestamped violation logs.

How It Works

The system leverages advanced **MediaPipe Face Mesh** and **OpenCV** to map the candidate's facial geometry in real-time.

- 1. Head Pose Tracking:** Analyzes head orientation to detect if the student is looking down at a phone or away from the monitor.
- 2. Eye Gaze (Concentration):** Calculates the position of the iris. Prolonged deviation triggers an anomaly state.
- 3. Automated Logging:** Violations are recorded instantly in the *Violation Log* and flagged for teacher review via Email API.

[Live Camera Mesh Feed]

Head Pose:
Normal

Eye Gaze:
Focused

[15:03:20] **ABSENCE DETECTED:**
No student found in front of the camera for over 3 seconds!

System Advantages

Zero Special Hardware

Runs efficiently on standard student laptops using built-in webcams. No

expensive eye-tracking equipment is required.

Scalable Monitoring

Allows a single teacher to reliably monitor hundreds of students simultaneously by relying on AI-filtered alert queues.

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FACIAL LANDMARKS

< 3s

ALERT LATENCY

100%

AUTOMATED